UNIT-1

1.Introduction

A **Computer Network** is a collection of two or more computers and other devices connected through wired or wireless communication links to exchange data, share resources, and communicate with each other.

2. Types of Computer Networks

Computer networks are classified based on the geographical area they cover.

Personal Area Network (PAN):

A **Personal Area Network (PAN)** is the smallest computer network that connects personal electronic devices within a short distance (usually **up to 10 metres**).

It is mainly used by one person to connect devices such as smartphones, laptops, smartwatches, and Bluetooth headsets.

Characteristics:

- Covers a short distance (up to 10 metres).
- Uses Bluetooth, USB, NFC, or Infrared.
- Low power consumption.
- Easy to install and configure.
- Low cost.

Example

A smartphone connected to Bluetooth earbuds and a smartwatch.

Local Area Network (LAN)

A **Local Area Network (LAN)** connects computers within a limited area such as a school, office, laboratory, or college campus.

Characteristics of LAN (Local Area Network)

- Covers a small geographical area (such as a room, office, school, or building).
- Provides high-speed data transfer (typically 100 Mbps to several Gbps).
- Uses Ethernet cables or Wi-Fi for communication.
- Low installation and maintenance cost compared to larger networks.
- Allows sharing of resources such as files, printers, and internet connections.

Example

A college computer laboratory.

Home Network

A **Home Network** is a small LAN created inside a house to connect multiple devices using a Wi-Fi route

Characteristics of Home Network

- Covers a single home or apartment.
- Connects devices such as computers, smartphones, TVs, printers, and gaming consoles.
- Uses Wi-Fi or Ethernet cables for communication.
- Provides internet access to all connected devices through a single router.
- Easy to install and manage.

Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** is a computer network that connects multiple Local Area Networks (LANs) within a city or metropolitan area. It covers a larger area than a LAN but a smaller area than a WAN.

Characteristics of Metropolitan Area Network

- Covers a city or metropolitan area (10–100 km).
- Connects multiple LANs.
- Provides high-speed communication.
- Usually uses optical fibre cables.
- Suitable for large organisations and government offices.

Example

A bank connecting all its branches in a city.

Wide Area Network (WAN)

A **Wide Area Network (WAN)** is a computer network that connects computers, Local Area Networks (LANs), and Metropolitan Area Networks (MANs) over a **large geographical area**, such as **countries or continents**.

Characteristics of WAN

- Covers a large geographical area, such as countries and continents.
- Connects LANs and MANs located in different places.
- Uses optical fibre, satellites, leased lines, and the Internet for communication.
- Supports communication over long distances.
- Provides global connectivity.

Example

A multinational company connecting offices in different countries.

3. Broadband Access Networks

A **Broadband Access Network** is a high-speed communication network that provides users with continuous Internet access. It connects **homes, offices, schools, and businesses** to the Internet using technologies such as **Digital Subscriber Line (DSL), Cable Modem, Fibre Optic, and Wireless Broadband**.

Unlike dial-up connections, broadband provides **high-speed, always-on Internet access**, allowing users to browse the web, stream videos, download files, and make video calls simultaneously.

Need for Broadband Access Networks

Broadband networks are required because they:

- Provide high-speed Internet access.
- Support multiple users and devices.
- Enable online education, video conferencing, cloud computing, and entertainment.
- Offer a stable and continuous Internet connection.

Types of Broadband Access Networks

1. Digital Subscriber Line (DSL)

DSL provides broadband Internet using existing **telephone lines** without interrupting voice calls.

Advantages

- Uses existing telephone infrastructure.
- Internet and telephone can work simultaneously.
- Easy to install.

Example

Many homes use DSL broadband provided through telephone lines.

2. Cable Modem

A cable modem provides Internet access through **cable television (CATV) networks**.

Advantages

- Faster than DSL.
- Suitable for video streaming.
- Easy installation.

Example

Cable TV operators providing broadband Internet.

3. Fibre Optic Broadband (FTTH – Fibre to the Home)

Fibre optic broadband uses **optical fibre cables** to transmit data as light signals, providing extremely high-speed Internet.

Advantages

- Very high speed.
- Reliable connection.
- Low signal loss.

Example

Home fibre connections provided by ISPs.

4. Wireless Broadband

Wireless broadband provides Internet access **without physical cables**, using technologies such as **Wi-Fi, 4G, and 5G**.

Advantages

- Easy to access anywhere within coverage.
- No cables required.
- Supports mobility.

Example

Mobile Internet using 4G or 5G.

4. Mobile and Wireless Access Networks

A **Mobile and Wireless Access Network** is a communication network that allows users to access the Internet and network services **without using physical cables**. It enables devices such as **smartphones, laptops, tablets, and smartwatches** to communicate using **radio waves** while allowing users to move from one place to another.

Wireless access networks use technologies such as **Wi-Fi, Bluetooth, 3G, 4G, and 5G** to provide Internet connectivity.

Need for Mobile and Wireless Access Networks

Mobile and wireless networks are used because they:

- Provide Internet access without cables.

- Allow users to connect while travelling.
- Support smartphones and portable devices.
- Enable communication anytime and anywhere.
- Reduce the need for wired infrastructure.

Components of a Wireless Access Network

1. Mobile Device

Devices such as smartphones, tablets, laptops, and smartwatches that access the network.

2. Access Point (Wi-Fi Router)

Provides wireless connectivity to nearby devices.

3. Base Station (Mobile Tower)

Connects mobile users to the cellular network.

4. Internet Service Provider (ISP)

Provides Internet connectivity to users.

Types of Mobile and Wireless Access Networks

1. Wi-Fi Network

Wi-Fi is a wireless networking technology that connects devices to the Internet using a wireless router.

Features

- Wireless Connectivity – No cables are required to connect devices.
- Easy Installation – Wi-Fi networks are simple and quick to set up.
- Mobility – Users can move freely within the Wi-Fi coverage area while staying connected.
- High-Speed Internet – Provides fast Internet access for browsing, downloading, and streaming.

2.Cellular Network (3G, 4G, 5G)

A cellular network provides Internet and voice communication through mobile towers.

Features

- Wide Coverage – Covers large areas such as cities, states, and even countries through mobile towers.
- Wireless Communication – Uses radio waves to transmit voice, data, and multimedia.
- Mobility – Users can move from one place to another while staying connected.

- Supports Voice and Data – Provides voice calls, SMS, Internet access, and multimedia services.

3. Bluetooth Network

Bluetooth is a short-range wireless communication technology used to connect personal devices.

Features

- Short-Range Communication-Bluetooth works within a short distance, usually up to 10 metres.
- Wireless Technology - It connects devices without using cables.
- Low Power Consumption -Bluetooth uses very little battery power, making it suitable for portable devices.
- Easy to Connect -Devices can be paired quickly and easily.

5. Content Provider Networks (CPN)

A **Content Provider Network (CPN)** is a network that delivers digital content such as **web pages, videos, images, music, software, cloud services, and online applications** to users quickly and efficiently.

A CPN stores copies of content on **multiple servers** located in different geographical locations. When a user requests content, it is delivered from the **nearest server**, reducing delay and improving performance.

Examples include **YouTube, Netflix, Google, Facebook, Amazon Prime Video, and Microsoft Azure**.

Need for Content Provider Networks

Content Provider Networks are required because:

- Millions of users access the same content simultaneously.
- Delivering content from one central server increases network traffic.
- Users expect fast loading of websites and videos.
- CPN reduces delay and improves user experience.

Components of a Content Provider Network

1. Main Server (Origin Server)

Stores the original copy of the content.

Example

YouTube stores original videos in its data centres.

2. Edge Servers

Edge servers store copies of frequently accessed content closer to users.

Example

A user in Hyderabad receives YouTube videos from the nearest edge server instead of a distant data centre.

3. Internet Backbone

Connects the main server, edge servers, and users.

4. Users

Users access websites, videos, applications, and cloud services through the Internet.

Working of a Content Provider Network

1. The content provider uploads content to the **main server**.
2. Copies of the content are stored on **multiple edge servers**.
3. When a user requests content, the request is sent to the **nearest edge server**.
4. The nearest server delivers the content quickly.
5. This reduces network congestion and improves download speed.

6. Transit Networks

A **Transit Network** is a **high-speed backbone network** that carries data traffic between different networks, Internet Service Providers (ISPs), data centres, and countries. It acts as a **bridge** that transfers data from one network to another, enabling communication across the Internet.

A transit network does not usually provide Internet access directly to end users. Instead, it **transports data between networks** quickly and efficiently.

Need for Transit Networks

Transit networks are essential because they:

- Connect different Internet Service Providers (ISPs).
- Transfer data over long distances.
- Reduce network congestion.
- Provide high-speed communication.
- Support global Internet connectivity.

Components of a Transit Network

1. Internet Backbone

The Internet backbone is the main high-speed network that carries Internet traffic.

Example

International optical fibre cable networks.

2. Internet Service Providers (ISPs)

ISPs connect users to the transit network.

Example

Jio, Airtel, BSNL, ACT Fibernet.

3. Routers

Routers direct data packets from one network to another.

4. Data Centres

Store websites, cloud services, and applications connected through transit networks.

Working of a Transit Network

1. A user sends a request through their local ISP.
2. The ISP forwards the request to a **transit network**.
3. The transit network carries the data through its high-speed backbone.
4. The data reaches the destination ISP.
5. The destination ISP delivers the data to the required user or server.

7. Enterprise Networks

An **Enterprise Network** is a computer network that connects all the computers, servers, departments, branch offices, employees, and other devices within an organisation. It enables secure communication, resource sharing, and efficient management of business operations.

Enterprise networks are commonly used by companies, banks, hospitals, universities, government organisations, and multinational corporations.

Components of an Enterprise Network

1. Computers and Workstations

Used by employees to perform daily tasks.

2. Servers

Store files, databases, applications, and email services.

3. Switches

Connect computers and other devices within the local network.

4. Routers

Connect the enterprise network to branch offices and the Internet.

5. Firewalls

Protect the network from cyber threats and unauthorised access.

6. Wireless Access Points

Provide Wi-Fi connectivity to laptops, smartphones, and tablets.

Working of an Enterprise Network

1. All computers and devices are connected through switches and routers.
2. A central server stores important business data.
3. Employees access files, applications, and printers through the network.
4. Branch offices communicate with the head office using secure WAN or VPN connections.
5. Firewalls and security systems protect the network from unauthorised access.

8. Network Technology from Local to Global

Network Technology from Local to Global refers to the evolution of computer networks from connecting devices within a **small area (Local)** to connecting millions of devices across the **entire world (Global)**. Different types of networks are used based on the geographical area they cover and the communication requirements.

The progression of network technology is:

PAN → LAN → MAN → WAN → Internet (Global Network)

1. Personal Area Network (PAN) – Local Level

A **Personal Area Network (PAN)** is the smallest computer network that connects personal electronic devices within a short distance (usually **up to 10 metres**).

It is mainly used by one person to connect devices such as smartphones, laptops, smartwatches, and Bluetooth headsets.

Characteristics:

- Covers a short distance (up to 10 metres).
- Uses Bluetooth, USB, NFC, or Infrared.
- Low power consumption.
- Easy to install and configure.
- Low cost.

Example

A smartphone connected to Bluetooth earbuds and a smartwatch.

Local Area Network (LAN)

A **Local Area Network (LAN)** connects computers within a limited area such as a school, office, laboratory, or college campus.

Characteristics of LAN (Local Area Network)

- Covers a small geographical area (such as a room, office, school, or building).
- Provides high-speed data transfer (typically 100 Mbps to several Gbps).
- Uses Ethernet cables or Wi-Fi for communication.
- Low installation and maintenance cost compared to larger networks.
- Allows sharing of resources such as files, printers, and internet connections.

Example

A college computer laboratory.

Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** is a computer network that connects multiple Local Area Networks (LANs) within a city or metropolitan area. It covers a larger area than a LAN but a smaller area than a WAN.

Characteristics of Metropolitan Area Network

- Covers a city or metropolitan area (10–100 km).
- Connects multiple LANs.
- Provides high-speed communication.
- Usually uses optical fibre cables.
- Suitable for large organisations and government offices.

Example

A bank connecting all its branches in a city.

Wide Area Network (WAN)

A **Wide Area Network (WAN)** is a computer network that connects computers, Local Area Networks (LANs), and Metropolitan Area Networks (MANs) over a **large geographical area**, such as **countries or continents**.

Characteristics of WAN

- Covers a large geographical area, such as countries and continents.
- Connects LANs and MANs located in different places.
- Uses optical fibre, satellites, leased lines, and the Internet for communication.
- Supports communication over long distances.
- Provides global connectivity.

Example

A multinational company connecting offices in different countries.

Global Network (Internet)

The **Internet** is the largest computer network in the world. It connects millions of computers and networks globally.

Characteristics

- Worldwide coverage
- Uses TCP/IP protocol
- Supports billions of users
- Available 24×7

Example

Google, YouTube, WhatsApp, Gmail, Amazon.

9. Personal Area Network (PAN)

A **Personal Area Network (PAN)** is the smallest computer network that connects personal electronic devices within a short distance (usually **up to 10 metres**).

It is mainly used by one person to connect devices such as smartphones, laptops, smartwatches, and Bluetooth headsets.

Characteristics:

- Covers a short distance (up to 10 metres).
- Uses Bluetooth, USB, NFC, or Infrared.
- Low power consumption.
- Easy to install and configure.
- Low cost.

Example

A smartphone connected to Bluetooth earbuds and a smartwatch.

Technologies Used in PAN

1. Bluetooth

- Most common PAN technology.
- Used for wireless file sharing and connecting accessories.
- Range: Up to **10 metres**.

Example

Connecting a smartphone to Bluetooth earbuds.

2. USB (Universal Serial Bus)

- Provides wired communication.
- Used to transfer data and charge devices.

Example

Connecting a smartphone to a laptop using a USB cable.

3 Infrared (IR)

- Uses infrared light for communication.
- Requires a direct line of sight between devices.

Example

TV remote controls.

4. Wi-Fi Direct

- Allows two devices to communicate directly without a Wi-Fi router.

Example

Sharing files between two smartphones.

10. Local Area Network (LAN)

A **Local Area Network (LAN)** connects computers within a limited area such as a school, office, laboratory, or college campus.

Characteristics of LAN (Local Area Network)

- Covers a small geographical area (such as a room, office, school, or building).
- Provides high-speed data transfer (typically 100 Mbps to several Gbps).
- Uses Ethernet cables or Wi-Fi for communication.
- Low installation and maintenance cost compared to larger networks.
- Allows sharing of resources such as files, printers, and internet connections.

Example

A college computer laboratory.

Components of LAN

1. Computers

Used by users to send and receive data.

Example

Desktop computers in a computer laboratory.

2. Network Switch

Connects multiple computers within the LAN and forwards data to the correct device.

3. Router

Connects the LAN to the Internet and enables communication with external networks.

4. Server

Stores files, databases, software, and applications for network users.

Working of LAN

1. Computers are connected using **Ethernet cables** or **Wi-Fi**.
2. A **switch** transfers data between connected devices.
3. A **router** connects the LAN to the Internet.
4. A **server** stores files and applications that can be accessed by all authorised users.
5. Users can share files, printers, and Internet connections through the network.

11.Home Networks

A **Home Network** is a small LAN created inside a house to connect multiple devices using a Wi-Fi route

Characteristics of Home Network

- Covers a single home or apartment.

- Connects devices such as computers, smartphones, TVs, printers, and gaming consoles.
- Uses Wi-Fi or Ethernet cables for communication.
- Provides internet access to all connected devices through a single router.
- Easy to install and manage.

Components of a Home Network

1. Modem

Connects the home to the Internet provided by the ISP.

2. Wi-Fi Router

Distributes the Internet connection to all devices. Enables wireless communication.

3. Computers and Laptops

Used for browsing, online classes, office work, and entertainment.

4. Smartphones and Tablets

Access the Internet, social media, and communication apps.

Working of a Home Network

1. The Internet Service Provider (ISP) provides an Internet connection through a modem.
2. The modem connects to a Wi-Fi router.
3. The router distributes the Internet connection to multiple devices.
4. Devices communicate with each other through the home network.
5. Users can share files, printers, and multimedia resources.

12. Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** is a computer network that connects multiple Local Area Networks (LANs) within a city or metropolitan area. It covers a larger area than a LAN but a smaller area than a WAN.

Characteristics of Metropolitan Area Network

- Covers a city or metropolitan area (10–100 km).
- Connects multiple LANs.
- Provides high-speed communication.
- Usually uses optical fibre cables.
- Suitable for large organisations and government offices.

Example

A bank connecting all its branches in a city.

Components of MAN

1. Local Area Networks (LANs)

Several LANs are connected to form a MAN.

Example

College campuses located in different parts of a city.

2. Routers

Routers connect different LANs and direct data between them.

3. Optical Fibre Cables

Used as the transmission medium because they provide high-speed communication.

4. Switches

Connect computers within each LAN.

5. Servers

Store shared files, databases, and applications.

Working of MAN

1. Different LANs in a city are connected using optical fibre cables.
2. A router transfers data between the connected LANs.
3. Users in different locations can access shared files, printers, databases, and applications.
4. The network provides fast and secure communication across the city.

13. Wide Area Network(WAN)

A **Wide Area Network (WAN)** is a computer network that connects computers, Local Area Networks (LANs), and Metropolitan Area Networks (MANs) over a **large geographical area**, such as **countries or continents**.

Characteristics of WAN

- Covers a large geographical area, such as countries and continents.
- Connects LANs and MANs located in different places.
- Uses optical fibre, satellites, leased lines, and the Internet for communication.
- Supports communication over long distances.
- Provides global connectivity.

Example

A multinational company connecting offices in different countries.

Components of WAN

1. Routers

Connect different LANs and direct data between networks.

2. Transmission Media

WAN uses:

- Optical Fibre Cables
- Leased Lines
- Microwave Links
- Satellite Communication
- Internet

3. Switches

Used to connect devices within local networks before they connect to the WAN.

4. Servers

Store business data, applications, and databases that can be accessed from different locations.

5. Internet Service Providers (ISPs)

Provide connectivity between networks and the Internet.

Working of WAN

1. Each office has its own LAN.
2. The LANs are connected using routers.
3. Routers communicate through leased lines, fibre optic cables, microwave links, satellites, or the Internet.
4. Data is transmitted from one location to another over the WAN.
5. Users can access shared files, databases, and applications regardless of their location.

14.Internetworks

An **Internetwork** is a collection of **two or more independent computer networks** that are connected together to communicate and exchange data. These networks may be **LANs, MANs, or WANs**, and they are interconnected using **routers and networking devices**.

Internetworking enables computers in different networks to communicate as if they are part of one large network.

The **Internet** is the **largest and best example of an internetwork**.

Components of an Internetwork

1. Local Area Networks (LANs)

Small networks that are connected to form a larger internetwork.

Example

Office networks in different buildings.

2. Routers

Connect different LANs and direct data between networks.

3. Transmission Media

WAN uses:

- Optical Fibre Cables
- Leased Lines
- Microwave Links
- Satellite Communication
- Internet

4. Switches

Used to connect devices within local networks before they connect to the WAN.

5. Servers

Store business data, applications, and databases that can be accessed from different locations.

6. Internet Service Providers (ISPs)

Provide connectivity between networks and the Internet.

Working of an Internetwork

1. A user sends data from one network.
2. The data reaches the local **router**.
3. The router forwards the data through the **Internet or another interconnected network**.
4. The destination router receives the data.
5. The destination network delivers the data to the required computer.

15. Network Protocols

A **Network Protocol** is a set of **rules and standards** that govern how data is transmitted, received, and interpreted between devices in a computer network.

Just as people use a common language to communicate, computers use **network protocols** to exchange information accurately and efficiently.

Need for Network Protocols

Network protocols are required because they:

- Establish communication between devices.
- Ensure correct transmission of data.
- Detect and recover from errors.
- Control the flow of data.

Types of Network Protocols

1. TCP (Transmission Control Protocol)

TCP is a **connection-oriented protocol** that ensures reliable and error-free data transmission.

Features

- Reliable communication
- Error detection and correction
- Data sequencing
- Flow control

Applications

- Email
- Web browsing
- File transfer

2. IP (Internet Protocol)

IP is responsible for **addressing and routing** data packets from the source to the destination.

Features

- Assigns IP addresses
- Routes packets
- Supports internetwork communication

Applications

- Internet communication
- Local area networks

3. HTTP (Hypertext Transfer Protocol)

HTTP is used for transferring **web pages** between web browsers and web servers.

Applications

- Websites
- Web browsers

Example

Opening a website using Google Chrome.

4. HTTPS (Hypertext Transfer Protocol Secure)

HTTPS is the secure version of HTTP that encrypts data during transmission.

Applications

- Online banking
- Shopping websites
- Secure websites

5. FTP (File Transfer Protocol)

FTP is used to transfer files between computers over a network.

Applications

- Uploading websites
- File sharing

6. SMTP (Simple Mail Transfer Protocol)

SMTP is used to send emails over the Internet.

Applications

- Gmail
- Outlook
- Yahoo Mail

7. POP3 (Post Office Protocol Version 3)

POP3 is used to download emails from a mail server to a computer.

8. IMAP (Internet Message Access Protocol)

IMAP allows users to access and manage emails directly on the mail server from multiple devices.

9. DNS (Domain Name System)

DNS converts **domain names** into **IP addresses**.

Example

Converts **www.google.com** into its IP address.

10. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses to devices connected to the network.

16. Design Goals

The Objectives of Computer Networks are the main goals or purposes of connecting two or more computers and devices. The primary objective is to enable communication, resource sharing, data sharing, and efficient use of computing resources.

1. Performance

Performance refers to how efficiently a computer network transfers data between devices. A high-performance network delivers data **quickly, accurately, and with minimum delay**.

Network performance depends on:

- **Bandwidth** – Amount of data that can be transmitted per second.
- **Throughput** – Actual amount of data successfully transferred.
- **Latency** – Time taken for data to travel from sender to receiver.
- **Response Time** – Time taken by the network to respond to a request.

A network with high bandwidth and low latency provides better performan

2. Reliability

Reliability is the ability of a network to provide **continuous and error-free communication** even when some network components fail.

A reliable network:

- Delivers data accurately.
- Recovers from failures quickly.
- Uses backup links and backup servers.
- Minimizes network downtime.

3. Scalability

Scalability is the ability of a network to **grow by adding more users, devices, or locations** without significantly reducing performance

A scalable network allows organizations to expand without replacing the entire network infrastructure.

Advantages

- Supports future expansion.
- Easy to add new devices.
- Reduces upgrade costs.
- Suitable for growing organizations.

4. Security

Security protects the network and its data from **unauthorized access, hacking, viruses, malware, and cyberattacks**.

Security is achieved using:

- Passwords
- Firewalls
- Encryption
- Antivirus software
- User authentication

5. Availability

Availability is the ability of a network to remain **operational and accessible whenever users need it**.

A highly available network:

- Works 24×7.
- Minimizes downtime.
- Uses backup equipment.
- Continues service even during hardware failures.

6. Cost Efficiency

Cost Efficiency means designing a network that provides **maximum performance with minimum installation, operation, and maintenance cost**.

A cost-efficient network:

- Shares resources.
- Reduces hardware costs.
- Minimizes maintenance expenses.

- Uses existing infrastructure whenever possible.

7. Flexibility

Flexibility is the ability of a network to **adapt easily to new technologies, devices, and user requirements**.

A flexible network can:

- Support new applications.
- Connect different types of devices.
- Upgrade hardware and software easily.
- Adapt to changing business needs.

17. Protocol Layering

Protocol Layering is a method of organizing network communication into **multiple layers**, where each layer performs a specific function and communicates with the layer above and below it.

Each layer uses a **protocol** (a set of communication rules) to exchange data with the corresponding layer on another computer.

Protocol layering simplifies network design by dividing complex communication into smaller and manageable tasks.

Need for Protocol Layering

Protocol layering is required because it:

- Simplifies network communication.
- Divides complex tasks into smaller parts.
- Makes troubleshooting easier.
- Allows independent development of each layer.
- Improves compatibility between different devices.

Functions of Each Layer

1. Application Layer

- Provides network services to user applications.
- Enables communication between applications and the network.

Protocol

- HTTP
- HTTPS
- FTP
- SMTP

2. Transport Layer

- Ensures reliable data delivery.
- Performs segmentation, flow control, and error recovery.

Protocol

- TCP
- UDP

3. Network Layer

- Determines the best path for data.
- Performs logical addressing using IP addresses.

Protocol

- IP
- ICMP

4. Data Link Layer

- Provides node-to-node communication.
- Performs error detection.
- Uses MAC addresses.

5. Physical Layer

- Transmits bits over cables or wireless media.
- Converts data into electrical, optical, or radio signals.

18. Connections and Reliability

Connections and Reliability are two important concepts in computer networks.

- A **connection** is the communication link established between two devices to exchange data.
- **Reliability** is the ability of the network to deliver data **accurately, completely, and without errors**.

Types of Connections

There are **two types of network connections**:

1. Connection-Oriented Communication
2. Connectionless Communication

1. Connection-Oriented Communication

In **connection-oriented communication**, a connection is established between the sender and receiver **before data transmission begins**.

After all data is transmitted, the connection is terminated.

This type of communication provides **high reliability**.

Working

1. Establish connection.
2. Send data.
3. Receive acknowledgment.
4. Terminate connection.

2. Connectionless Communication

In **connectionless communication**, data is transmitted **without establishing a connection** between sender and receiver.

Each packet is sent independently.

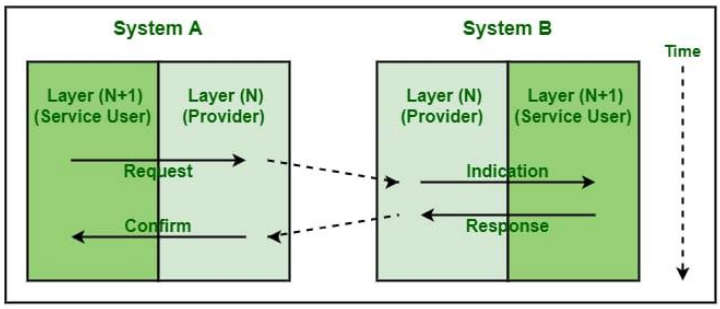
It is faster but less reliable.

Working

1. Sender sends packets.
2. Packets travel independently.
3. Receiver accepts available packets.
4. Lost packets are not retransmitted.

19. Service Primitives

A **Service Primitive** is a **basic operation or command** used by one layer of the network to request services from the layer below it.



Primitives for Connection between Peer Protocol Entities

Types of Service Primitives

There are **four types of service primitives**:

1. **REQUEST**
2. **INDICATION**
3. **RESPONSE**
4. **CONFIRM**

1. REQUEST Primitive

A **REQUEST** primitive is sent by the **service user** to the **service provider** to request a service.

The upper layer asks the lower layer to perform a specific task.

Example

A web browser requests the Transport Layer to establish a connection before sending data.

2. INDICATION Primitive

An **INDICATION** primitive is sent by the **service provider** to inform the **service user** that a request has been received or an event has occurred.

The lower layer informs the upper layer that a service request has arrived.

Example

The receiver's Transport Layer informs the Application Layer that another computer wants to establish a connection.

3. RESPONSE Primitive

A **RESPONSE** primitive is sent by the **service user** after receiving an indication to accept or reject the request.

The receiving application decides whether to accept or reject the connection request.

Example

The server accepts the client's request to establish a connection.

4. CONFIRM Primitive

A **CONFIRM** primitive is sent by the **service provider** to inform the **service user** that the requested service has been completed successfully.

The lower layer confirms that the requested operation has been completed.

Example

The Transport Layer informs the Application Layer that the connection has been established successfully.

20. Relationship of Services to Protocols

In computer networks, **services** and **protocols** are closely related, but they are **not the same**.

- A **service** defines **what a layer provides** to the layer above it.
- A **protocol** defines **how two corresponding layers in different computers communicate** to provide that service.

Service

A Service is the set of functions or facilities that one layer offers to the layer immediately above it.

Characteristics of a Service

- Defines **what** the layer provides.
- Independent of implementation.
- Used by the upper layer.
- Hidden from the user.

Example

The **Transport Layer** provides reliable data transfer to the **Application Layer**.

Protocol

A **protocol** is a set of rules that governs communication between corresponding layers on different computers.

Characteristics of a Protocol

- Defines **how** communication occurs.
- Specifies message format.
- Specifies timing.
- Specifies error handling.
- Specifies flow control.

Example

TCP is a protocol used by the Transport Layer for reliable communication.

Relationship Between Services and Protocols

Services and protocols work together.

Working

1. The upper layer requests a service.
2. The lower layer provides the requested service.
3. The lower layer communicates with the corresponding layer on another computer using a protocol.
4. The protocol follows predefined rules to exchange data.
5. The requested service is completed successfully.

21. Reference Models

A **Reference Model** is a **conceptual framework** that describes how data is transmitted from one computer to another over a network. It divides the communication process into **multiple layers**, where each layer performs a specific function.

Reference models help in **designing, developing, and understanding computer networks**. They provide a standard structure so that devices from different manufacturers can communicate with each other.

The two most important reference models are:

1. **OSI (Open Systems Interconnection) Reference Model**
2. **TCP/IP Reference Model**

1. OSI (Open Systems Interconnection) Reference Model

The **OSI Reference Model** was developed by the **International Organization for Standardization (ISO)** in **1984**. It divides network communication into **seven layers**, with each layer performing a specific function.

Functions of OSI Layers

1. Physical Layer

- Transmits bits over the communication medium.
- Uses cables, fiber optics, and wireless signals.

Examples: Ethernet cable, Fiber Optic, Bluetooth.

2. Data Link Layer

- Provides node-to-node communication.
- Performs framing, error detection, and MAC addressing.

Examples: Ethernet, Wi-Fi.

3. Network Layer

- Performs logical addressing and routing.
- Finds the best path for data.

Protocol: IP

4. Transport Layer

- Provides reliable data delivery.
- Performs segmentation, flow control, and error control.

Protocols: TCP, UDP

5. Session Layer

- Establishes, manages, and terminates communication sessions.

Examples: NetBIOS, RPC.

7. Application Layer

- Provides network services directly to users.

Protocols: HTTP, HTTPS, FTP, SMTP, DNS.

2. TCP/IP Reference Model

The **TCP/IP Reference Model** was developed by the **U.S. Department of Defense (DoD)** for communication over the Internet. It is the practical model used in modern computer networks.

It consists of **four layers**.

- Functions of TCP/IP Layers

1. Application Layer

- Provides services to user applications.
- Combines the functions of the OSI Application, Presentation, and Session layers.

Protocols: HTTP, HTTPS, FTP, SMTP, DNS.

2. Transport Layer

- Provides end-to-end communication.
- Ensures reliable data transfer.

Protocols: TCP, UDP.

3. Internet Layer

- Performs logical addressing and routing.

Protocol: IP

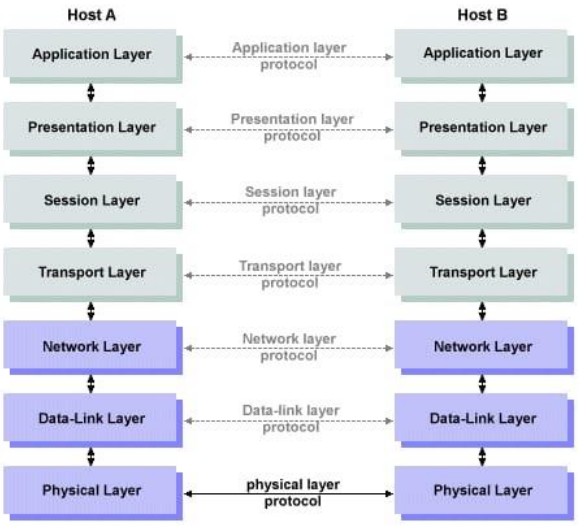
4. Network Access Layer

- Handles data transmission over the physical network.
- Includes functions of the OSI Physical and Data Link layers.

Examples: Ethernet, Wi-Fi.

22. OSI Reference Model

The OSI (Open Systems Interconnection) Reference Model is a standard networking model developed by the International Organization for Standardization (ISO) in 1984. It explains how data is transmitted from one computer to another through a network by dividing the communication process into seven layers.



Functions of the Seven Layers

1. Physical Layer (Layer 1)

The Physical Layer is responsible for transmitting **raw bits (0s and 1s)** through the communication medium.

Responsibilities

- Transmits bits.
- Defines cables and connectors.
- Converts data into electrical, optical, or radio signals.

Examples

- UTP Cable
- Fiber Optic Cable
- Bluetooth
- USB

2. Data Link Layer (Layer 2)

The Data Link Layer provides **node-to-node communication** and ensures error-free transmission over a physical link.

Responsibilities

- Framing
- Error detection
- Flow control
- MAC addressing

Protocols

- Ethernet (IEEE 802.3)
- Wi-Fi (IEEE 802.11)

3. Network Layer (Layer 3)

The Network Layer is responsible for **routing** and **logical addressing**.

Responsibilities

- IP addressing
- Routing
- Packet forwarding

Protocols

- IP
- ICMP
- OSPF

4. Transport Layer (Layer 4)

The Transport Layer provides **end-to-end communication** and ensures reliable data delivery.

Responsibilities

- Segmentation
- Error control
- Flow control

- Reliable delivery

Protocols

- TCP
- UDP

5. Session Layer (Layer 5)

The Session Layer establishes, manages, and terminates communication sessions between applications.

Responsibilities

- Session establishment
- Session maintenance
- Session termination

Protocols

- NetBIOS
- RPC
- SIP

6. Presentation Layer (Layer 6)

The Presentation Layer converts data into a format that both computers can understand.

Responsibilities

- Data translation
- Encryption
- Decryption
- Compression

Protocols/Formats

- SSL/TLS
- JPEG
- PNG
- MPEG

7. Application Layer (Layer 7)

The Application Layer provides network services directly to user applications.

Responsibilities

- Web browsing
- Email

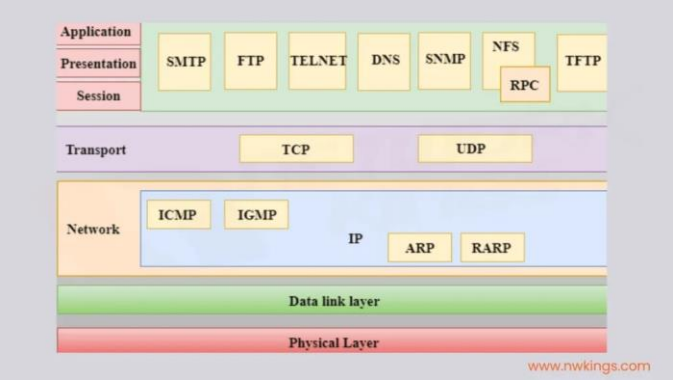
- File transfer
- Remote login

Protocols

- HTTP
- HTTPS
- FTP
- SMTP

23. TCP/IP Reference Model

The TCP/IP model divides network communication into **four layers**, with each layer performing a specific function. It is the **foundation of the modern Internet** and is used worldwide.



Layers of the TCP/IP Model

The TCP/IP model consists of **four layers**:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

1. Application Layer

The Application Layer provides **network services directly to user applications**. It combines the functions of the **Application, Presentation, and Session layers** of the OSI model.

Responsibilities

- Web browsing
- Email communication
- File transfer
- Remote login
- Name resolution

Protocols

- HTTP
- HTTPS
- FTP
- SMTP
- POP3
- IMAP

2. Transport Layer

The Transport Layer provides **end-to-end communication** and ensures reliable or fast data delivery.

Responsibilities

- Segmentation
- Error control
- Flow control
- Reliable delivery
- Port addressing

Protocols

- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)

3. Internet Layer

The Internet Layer is responsible for **logical addressing and routing**. It determines the best path for data to travel from sender to receiver.

Responsibilities

- IP Addressing
- Routing
- Packet forwarding
- Fragmentation

Protocols

- IP (IPv4, IPv6)
- ICMP
- IGMP
- ARP

4. Network Access Layer

The Network Access Layer is responsible for transmitting data over the physical network. It combines the functions of the **Physical Layer** and **Data Link Layer** of the OSI model.

Responsibilities

- Framing
- Physical addressing (MAC)
- Error detection
- Bit transmission
- Media access control

Protocols/Technologies

- Ethernet
- Wi-Fi (IEEE 802.11)
- PPP
- Bluetooth
- Fiber Optic
- DSL

24. Critique of the OSI Model and Protocols

A **critique** is a detailed evaluation of the strengths and weaknesses of a system.

The **OSI (Open Systems Interconnection) Model** is an important reference model for computer networks. Although it provides a standard framework for network communication, it also has several limitations.

Advantages (Strengths) of the OSI Model

1. Standardization

The OSI model provides an international standard for network communication.

2. Layered Architecture

The communication process is divided into seven layers, and each layer performs a specific function.

3. Easy Troubleshooting

Since every layer performs a separate function, network problems can be identified layer by layer

4. Modular Design

Each layer works independently. Changes in one layer usually do not affect the other layers.

Disadvantages (Weaknesses) of the OSI Model

1. Complex Model

The OSI model has **seven layers**, making it difficult for beginners to understand and implement.

2. Layers are Sometimes Redundant

Some layers perform similar or overlapping functions.

3. Rarely Implemented Exactly

Most practical networks use the **TCP/IP model**, not the OSI model.

The OSI model is mainly used for learning and reference purposes.

4. Slow Adoption

OSI protocols were developed slowly and became available after TCP/IP was already widely accepted.

25. Critique of the TCP/IP Model and Protocols

A critique is the evaluation of the **advantages (strengths)** and **disadvantages (weaknesses)** of a system.

The **TCP/IP (Transmission Control Protocol/Internet Protocol) Model** is the most widely used networking model and forms the basis of the Internet. It provides reliable communication between different types of networks. However, despite its popularity, it has some limitations. Similarly, the **TCP/IP protocols** have both strengths and weaknesses.

Advantages (Strengths) of the TCP/IP Model

1. Simple Architecture

The TCP/IP model has only **four layers**, making it simpler than the seven-layer OSI model.

2. Widely Used

The TCP/IP model is used worldwide for Internet communication.

3. Reliable Communication

The **TCP protocol** provides reliable communication by using acknowledgments, error detection, retransmission, and flow control.

4. Scalable

The TCP/IP model supports communication in both small and large networks

Disadvantages (Weaknesses) of the TCP/IP Model

1. No Separate Presentation Layer

The TCP/IP model does not have a dedicated Presentation Layer.

Functions such as **encryption, compression, and data translation** are handled by the Application Layer.

2. No Separate Session Layer

Session management is also included in the Application Layer.

This makes the Application Layer more complex.

3. Protocol Dependent

The model is built specifically around TCP/IP protocols.

Replacing these protocols is not easy.

4. Does Not Clearly Define Services

The TCP/IP model does not clearly separate **services, interfaces, and protocols**, unlike the OSI model.

26. Difference between OSI & TCP/IP Model

OSI Reference Model	TCP/IP Reference Model
Developed by ISO (International Organization for Standardization).	Developed by DARPA (Defense Advanced Research Projects Agency).
Has 7 layers.	Has 4 layers.
It is a theoretical (reference) model used for learning and designing networks.	It is a practical model used for real Internet communication.
The model was developed first, and then protocols were designed based on it.	Protocols were developed first, and the model was created later.
Has separate Presentation and Session layers.	Does not have separate Presentation and Session layers; their functions are included in the Application layer.
Has separate Physical and Data Link layers.	Combines the Physical and Data Link layers into a single Network Access layer.
Clearly separates services, interfaces, and protocols.	Does not clearly distinguish between services, interfaces, and protocols.
The Network layer supports both connection-oriented and connectionless communication.	The Internet layer mainly provides connectionless communication using IP.
Mainly used for education, network design, and understanding networking concepts.	Widely used in real-world networks and the Internet.
Less commonly implemented as a complete protocol suite.	The most widely implemented networking model, forming the basis of the Internet.